

COL7(6)83 Assignment 1

Due date: 7 September, 2025

This assignment deals with basic image operations, intensity transformations, spatial filtering, and colour image processing. All tasks have to be implemented yourself by working with the array of image pixels, unless otherwise specified in the problem. Most of the tasks are about working with intensity images; if you are starting with a colour image, obtain the intensity by averaging the first 3 components.

The assignment is in two parts. Part 1 is being released on 14 August. Part 2 will be released in a few days.

PART 1

Suppose you want to implement one of those “document scanner” apps that let you take a photo of a paper document and enhance it to look more like a scanned image. Take one such photo yourself, making sure all 4 corners of the page are visible in the image, and use it as a test input. When taking the photo, tilt the camera slightly so the page is not perfectly vertical.

1. The first thing to do is a geometric transformation to straighten the document. Pick the coordinates of the four corners manually (you may use a built-in function for picking points, e.g. Matlab/Matplotlib's `ginput()`).
 - a. Using these coordinates, compute a rotation that tries to make this quadrilateral as vertically aligned as possible, then create a transformed image which contains the entire document and as little of the background as possible. Discuss the details of the procedure in the text of your report.
 - b. Show results of the transformation using both nearest neighbour interpolation and bilinear interpolation. Include images zooming in to a small part of the output image, so that the difference between the two methods is clearly visible.
2. Typically, a photograph of a printed page does not have high enough contrast. Implement a few ways to increase the contrast such that the paper becomes (nearly) white and the text becomes (nearly) black:
 - a. Perform thresholding with a manually chosen intensity threshold.

- b. Perform an intensity transformation of the form $T(r) = ar + b$. First, choose the parameters manually to get the best result for your image. Then, think of a way to automatically choose the parameters based on the image histogram, implement it, and discuss how it works in your report.
 - c. Perform histogram equalization, using a built-in function if you wish. Discuss whether it works well, and why or why not.
3. Let's add a watermark to the corner of the processed image, to encourage the user to buy the paid version of our hypothetical app :)
 - a. Obtain an image of the IITD logo from <https://internal.iitd.ac.in/?q=content/logo-library>. The logos in the Large category are numbered 19-24; use logo number $(\text{sum of your entry nos.} \bmod 5) + 19$. Using thresholding, obtain a mask image which distinguishes the white background from the logo itself.
 - b. Resize the logo image (using either interpolation scheme) so its width is 20% that of the document image; do the same for the mask image. Use some image arithmetic to draw the logo with 50% transparency at the bottom right of the document, i.e. for each pixel that is not part of the white background, the output intensity should be the average of the document intensity and the logo intensity. Express this operation as a formula involving the document d , logo l , and mask m in your report (assuming l and m are already transformed to align with the corner of d).

Now, we forget about our document scanner and consider a different problem: working with images having very high intensity resolution, such as high dynamic range images. You can download some such images from [Paul Debevec's webpage](#) about HDR image acquisition (scroll down to the "Radiance Maps" section).

4. The principle of histogram equalization is equally applicable to images with arbitrarily high intensity resolution, but the usual algorithm based on building a table of size L may no longer be practical.
 - a. Design an algorithm that performs histogram equalization even in the absence of quantization, i.e. there are a finite number of pixels, but each pixel's intensity is an arbitrary real number (though there can be multiple pixels with the same intensity), and the goal is to distribute the intensities uniformly over a target range $[a, b]$. Explain the method in your report.
 - b. Demonstrate your method on one of the HDR images above, mapping the intensities to the range $[0, 255]$.

- c. Prove mathematically that if a strictly increasing intensity transformation $T(r)$ is applied to the intensities, the result of histogram equalization afterwards is identical to histogram equalization of the original image. In particular, performing histogram equalization twice has no additional effect.

PART 2

The aim of the following two tasks is to compare the theoretical properties of linear spatial filtering with its practical behaviour on digital images, in the presence of effects like quantization and gamma correction.

5. Implement a convolution function that takes a floating-point kernel w , an 8-bit intensity image f , and an optional 8-bit scalar c , and returns the convolution $g = w * f + c$ as an 8-bit image. The intensity of each pixel $g(x, y)$ should be computed with sufficient precision, say in floating-point, and then saved as an 8-bit value in $[0, 255]$. The additive constant c is there to help deal with possible negative values in $w * f$. You may use any padding scheme, but specify which one in your report.
 - a. Demonstrate your function by computing the Laplacian $\nabla^2 f$ of an image of your choice by convolving it with a 3×3 kernel. Use $c = 128$.
 - b. Perform Gaussian filtering with a user-specified standard deviation σ in two ways: first, by using your convolution function naively with a 2D kernel; second, by exploiting the separability of the Gaussian function. Show both results, and report the compute times of the two approaches for a sufficiently large σ such that there is an appreciable difference.
 - c. It is sometimes desired to compute the Laplacian of a Gaussian-filtered image, $l * g * f$ where l is the 3×3 Laplacian kernel and g is a Gaussian kernel. Using a moderate value of σ , compute this in three ways: as $l * (g * f)$, as $g * (l * f)$, and as $(l * g) * f$, exploiting the separability of g wherever possible. Show all three results. Mathematically, all three should be identical, but in practice, you should observe some significant differences. In your report, explain why these occur.
6. Suppose we have an image f , to which we want to apply an intensity transformation function T and an arbitrary convolution kernel w (not necessarily summing to 1). Then, we have the option of transforming intensities before convolution, $w * T(f)$, or after, $T(w * f)$.
 - a. Suppose $T(r) = ar + b$ is a linear polynomial. Is there an intensity transformation T' which, when applied after convolution, gives the same results as T applied before convolution, i.e. $T'(w * f) = w * T(f)$? If yes, derive the formula for T' . If not, give a counterexample.

- b. Let us consider the case of nonlinear intensity transformations. In particular, use the HDR image you used in problem 4, and take T to be a gamma transformation followed by rescaling to $[0, 255]$; choose an appropriate value of $\gamma \ll 1$ such that intensity variations in both very dark and very bright regions can be seen adequately. Let w be a disk of radius r , i.e. $w(x, y) = c$ iff $x^2 + y^2 < r^2$, with c chosen so the kernel weights sum to 1. Compare the results of gamma transformation before convolution vs. after. Which result is more representative of an image taken with an out-of-focus camera?

An appropriate choice of colour space may make it easier to identify and manipulate coloured objects in an image, as we will see in the next task.

7. Take an image containing objects of many different colours, e.g. a bowl of assorted fruits, or a group of people wearing colourful clothes. Choose one of the coloured objects as the one you wish to modify; we will try to change it to a different colour.
- Convert the image into the HSI colour space and display the H, S, and I channels as intensity images.
 - Determine the set of all pixels lying on the desired object; you will represent this as a “mask” image whose value is 1 for pixels on the object and 0 elsewhere. To identify such pixels, perform *colour slicing* as described in Ch. 6.5 of the textbook: pick a seed pixel (x_s, y_s) on the object, and use a cuboid centered at its colour vector $\mathbf{c}_s = \mathbf{f}(x_s, y_s)$ to identify other pixels of similar colours. Try colour slicing in RGB and in HSI and show the resulting mask in both cases, choosing the size and shape of the cuboid to select as much of the object as possible while including none (or very little) of the background.
 - If you are given a target colour $\mathbf{c}_t$ in HSI, the object can be transformed to this colour by constructing a colour transformation $\mathbf{T} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $\mathbf{T}(\mathbf{c}_s) = \mathbf{c}_t$, and applying it to all pixels that are 1 in the mask image. Discuss in your report how this transformation function should be designed for realistic results, and show the outcome for a few different choices of $\mathbf{c}_t$. **Hint:** In the scalar case, one could choose an additive approach $T(c) = c + (c_t - c_s)$ or a multiplicative one $T(c) = (c_t/c_s)c$. Which choice is appropriate for the H, S, and I components?

SUBMISSION

You will submit a zip file that contains (i) all your code for the assignment, (ii) a PDF report that includes your results for each of the assignment problems. In the text of the report, also provide all relevant information for each part.

All images should ideally be saved in PNG format, which is lossless and so does not cause any information loss (other than quantization if the intensities are not already in 8-bit format). JPEG is

permitted if your submission PDF is becoming too big for the upload limit.

Your assignment should be submitted on Moodle (new) before midnight on the due date. Only one person in a group needs to submit. Late days are counted with a quantization of 0.5 days: if you cannot finish the assignment by midnight, get some sleep and submit by noon the following day.

Separately, each of you must individually submit a short response in a separate form regarding how much you and your partner contributed to the work in this assignment. Your assignment marks will not be released until you submit the form.